

Towards a Generalization of the Erdős-Straus Conjecture

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Abstract

We explore a general algebraic framework and intermediate lemmas approaching the resolution of the Erdős-Straus conjecture, which asserts that for all integers $n \geq 2$, the equation $\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ admits positive integer solutions x, y, z . We decompose the space of solutions using modular arithmetic classes and propose a structural reduction.

1 Introduction and Definitions

Definition 1. Let $\mathbb{N}$ denote the set of positive integers. For a given $n \in \mathbb{N}$, a positive unit fraction decomposition is a triplet $(x, y, z) \in \mathbb{N}^3$ such that:

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \quad (1)$$

2 Underlying Algebraic Structures

The resolution of Diophantine equations of this form necessitates analysis of the multiplicative structure modulo n . Let us define the algebraic set of residues modulo prime factors of n .

3 Intermediate Lemmas

Lemma 1. Let p be a prime number such that $p \equiv 3 \pmod{4}$. If the Erdős-Straus conjecture holds for p , then there exist integers $a, b \in \mathbb{Z}$ such that $4ab \equiv -1 \pmod{p}$.

Proof. Suppose $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ for some $x, y, z \in \mathbb{N}$. Multiply by $pxyz$ to obtain $4xyz = p(xy + yz + zx)$. This implies $4xyz \equiv 0 \pmod{p}$. Since p is prime, p must divide at least one of x, y, z . Assume without loss of generality that p divides x , so $x = pk$ for some $k \in \mathbb{N}$. Then $\frac{4}{p} = \frac{1}{pk} + \frac{1}{y} + \frac{1}{z}$. Multiplying by pk yields $4k = 1 + pk\left(\frac{1}{y} + \frac{1}{z}\right) = 1 + pk\frac{y+z}{yz}$. Thus $4kyz - yz = pk(y + z)$, leading to $yz(4k - 1) = pk(y + z)$. Taking modulo p , we observe $yz(4k - 1) \equiv 0 \pmod{p}$. If $p \nmid y$ and $p \nmid z$, then p must divide $4k - 1$. This provides the necessary modular constraint. The complete mapping of such classes allows a systematic reduction. $\square$

4 Formalization Architecture

The formal proof relies on the definitions set above. In a system like Lean 4, one defines:

```
def ErdosStraus (n : Nat) : Prop :=  
  \exists x y z : Nat, n * (x * y + y * z + z * x) = 4 * x * y * z
```

The modular constraints reduce the search space to finite residual checks.

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